COMPUTERS AND BIOMEDICAL RESEARCH **18**, 46–61 (1985)

Stepwise Variable Selection and Maximum Likelihood Estimation of Smoothing Factors of Kernel Functions for Nonparametric Discriminant Functions Evaluated by Different Criteria

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Received March 1, 1984

Kernel functions are used to estimate the probability density functions of variables for nonparametric discriminant analysis. In connection with stepwise variable identification a stepwise maximum likelihood estimation procedure for the estimation of smoothing factors of the kernel functions is developed. This procedure allows a step-by-step estimation of smoothing factors for every variable which is considered to be added to the model or which is examined to substitute a variable in a model. Different criteria for model evaluation in stepwise discriminant analysis are discussed. Beside criteria, like distance and dependence functions and the error and nonerror rate, a criterion which considers the ratio of probability densities of different classes at point x is proposed for stepwise variable identification. An application of the procedures described in this study to a medical decision problem shows the importance of stepwise parameter estimation of kernel functions for nonparametric discriminant analysis and the role of different model evaluation criteria for the selection of the best subset of variables. © 1985 Academic Press, Inc.

INTRODUCTION

The objective of discriminant analysis is to characterize different populations by a subset of variables which are selected out of a generally large number of variables according to an evaluation criterion. The variables are the input into a multivariate discrimination function which assigns an observation to one of a finite number of classes according to some decision rule. There is no unique criterion for the evaluation of the quality of a discriminant function and different evaluation criteria may lead to different subsets of variables which form the optimal discriminant function. Usually, the variables are selected in stepwise procedures to identify a class of at least suboptimal subsets of variables according to the evaluation criterion. The type of the discriminant function, which is used for the characterization of different classes, depends on the properties of the data which are analyzed.

0010-4809/85 $3.00

For many practical applications (*1*) in the last years Fisher's linear discriminant function (*2*) has been used. This linear function is a special form of a multiple regression problem. The main advantage of the linear discriminant function is that its model parameters can be estimated very easily. The disadvantages of this function are the following necessary conditions, which restrict its application for many practical purposes and which have to be fulfilled to achieve optimal classification results:

(i) multinormal distribution of the variables
(ii) identical variance–covariance matrices for all classes.

If one or both conditions are not fulfilled, the application of the linear discriminant functions can be very misleading (*3*). Simulation studies have shown that the linear discriminant function is rather robust when the distributions of the variables of different classes do not deviate markedly from normality and if they are similar in shape and have identical variance–covariance matrices (*3*, *4*). The linear discriminant function is very sensitive to unequal variance–covariance matrices also when the variables are normally distributed. In the case of larger sample sizes, quadratic discriminant functions show rather good results (*5*, *6*).

Considering these restrictions for practical applications of the linear and also of the quadratic discriminant function, it is necessary to search for procedures which are independent of such restrictions (*7*). One procedure, which has been developed in the last years (*5*, *8*, *9*) is a nonparametric discriminant analysis. This procedure is based on the estimation of probability density functions of classes by kernel functions. The main problems for an application of nonparametric discriminant analysis are the choice of the appropriate shape of kernel functions and the determination of smoothing factors for kernel functions. The estimation of appropriate smoothing factors needs much computing time and therefore nonparametric discriminant analysis has not been applied very frequently to practical problems (*10*, *11*). Theoretical investigations using simulated data have shown that nonparametric discriminant analysis in the form proposed by Hermans *et al.* (*8*) is superior to linear and quadratic discriminant analysis, if the conditions: multinormal distribution and identical variance–covariance matrices of the classes are not fulfilled. Only if these conditions are fulfilled, the linear discriminant function is superior to nonparametric discriminant functions with respect to their discriminatory ability (*6*).

Because of the high demand for computing time for the estimation of smoothing factors, rather strong simplifications of the kernel functions have been assumed, like the assumption of only one constant smoothing factor for all variables (*8*, *9*). For this study, a stepwise variable selection procedure was combined with a stepwise estimation of the smoothing factor. This procedure allows an adaptive estimation of smoothing factors for the variables which are selected.

The quality of a discriminant function depends on the number of variables which are used and therefore stepwise procedures are an important tool to reduce the number of variables without substantial loss of the discriminatory power. A common problem to all stepwise procedures is the choice of the evaluation criterion. In this study, different evaluation criteria are computed simultaneously for one model. It can be shown that different evaluation criteria may lead to different "optimal" subsets of variables.

The methodology described in this paper will be illustrated by an application to medical decision making. Nonparametric stepwise discriminant analysis is used to determine possible biochemical risk factors for stroke. Like the example demonstrated in this study nearly all medical decision problems are multifactorial with highly correlated variables. With increasing number of variables there is an increasing demand for procedures which summarize the information content of variables with different distributions and which assist medical diagnosis.

Estimation of Jackknife Probabilities of Nonparametric Discriminant Functions

Nonparametric discriminant analysis is based on the estimation of probability density functions $f(x/d_i;H)$ of classes d_i, $i = 1, \ldots, m$. Given n_i independent sample points

$$x_{i1}, \ldots, x_{in_i}$$

of class d_i, with

$$x'_{ij} = (x_{ij1}, \ldots, x_{ijq})$$

$$q \ldots \text{number of variables},$$

the probability density function of class d_i at point x can be estimated by kernel functions $K(x,x_{ij};H)$ (*9*):

$$\hat{f}_K(x/d_i;H) = \frac{1}{n_i} \sum_{j=1}^{n_i} K(x,x_{ij};H). \qquad [1]$$

H is $q \times q$ matrix of smoothing factors of the kernels. It is assumed that H is constant for all classes d_i, $i = 1, \ldots, m$. The problems of estimation of $f(x|d_i;H)$, $i = 1, \ldots, m$ are:

(i) the choice of the shape of the kernel functions $K(x,x_{ij};H)$
(ii) the determination of the elements of the $q \times q$ matrix H.

The shape of the kernels is not so important, but the values of the smoothing factors are (*12, 13*). Generally any function $K(x,x_{ij};H)$, which fulfills the properties of a probability density function may be used as a kernel function. The elements h_{lk}^2, $l = 1, \ldots, q$, $k = 1, \ldots, q$ of the matrix H represent the kernel width and determine the smoothness of the estimated probability density function. In this study, only continuous variables are considered and multinormal kernels with variance–covariance matrix H are chosen.

$$K(x,x_{ij};H) = \frac{1}{(2\pi)^{q/2}|H|^{1/2}} \exp[-1/2(x - x_{ij})'H^{-1}(x - x_{ij})]. \quad [2]$$

To reduce computational work the matrix H in Eq. [2] is assumed to be a $q \times q$ diagonal matrix.

The estimated probability density function at point x for class d_i, $i = 1, \ldots, m$ is

$$\hat{f}_K(x/d_i;H) = \frac{1}{n_i} \frac{1}{(2\pi)^{q/2} h_{11} \ldots h_{qq}} \sum_{j=1}^{n_i} \prod_{l=1}^{q} \exp\left[-\frac{1}{2}\left(\frac{x_l - x_{ijl}}{h_{ll}}\right)^2\right] \quad [3]$$

if multinormal uncorrelated kernel functions are used. A maximum likelihood criterion is used to estimate the elements of matrix H:

$$L(H) = \prod_{i=1}^{m} \prod_{j=1}^{n_i} \hat{f}_K(x_{ij}/d_i;H) \rightarrow \max. \quad [4]$$

A simultaneous solution of Eq. [4] for all elements of H is extremely complicated and therefore only one constant smoothing factor $h_{ll} = h_c$, $l = 1, \ldots, q$ for all variables is assumed (*8*, *14*).

Considering Eq. [1] the value of the probability density function at point x_{ij} depends on x_{ij}. This may lead to extreme values of the smoothing factors (*10*, *15*). A successful method, which was applied already by Lachenbruch (*16*) and was further developed in the last few years (*17*) is the leaving-one-out or jackknife method, which is used to estimate the probability density function:

$$\hat{f}_{K,J}(x_{ir}/d_i;H) = \frac{1}{n_i - 1} \sum_{\substack{j=1 \\ j \neq r}}^{n_i} K(x_{ir},x_{ij};H). \quad [5]$$

These jackknife probability densities are used in Eq. [4] for the estimation of the smoothing factors.

The assumption of only one constant smoothing factor h_c for all variables seems to be a rather strong simplification. Considering the maximum likelihood estimated smoothing factors of single variables which have been standardized to variance 1, it can be shown that different variables can have very different smoothing factors (Table II). To avoid this problem of a constant smoothing factor for all variables in multivariate discriminant functions ($q > 1$) and to avoid also the problems of simultaneous solution of the likelihood function [4], the estimation of smoothing factors for single variables was combined with the stepwise variable identification procedure.

Stepwise Variable Selection and Maximum Likelihood Estimation of Smoothing Factors

The problems of variable selection in discriminant analysis are very similar to variable selection in pattern recognition (*18*) or in multiple regression (*19*, *20*). The aim is to identify a "best" subset of p variables out of q variables according to some evaluation criterion. The aim of variable selection in stepwise discriminant analysis is to identify those variables which contain most of the relevant

TABLE I

INTERQUARTILE RANGES (I_{25}–I_{75}) AND MEDIANS (x) OF q = 15 VARIABLES OF THE RISK (n = 45) AND CONTROL (n = 31) GROUP[a]

		Risk			Control			
$X(k)$		I_{25}	x	I_{75}	I_{25}	x	I_{75}	z
1	Lecithin (Lec)	90.54	107.59	133.55	122.41	134.49	150.61	3.11
2	Free cholesterol (FC)	34.05	43.07	56.20	41.65	49.20	55.74	1.23
3	High density lipoproteine cholesterol (HDL-C)	25.71	29.30	36.30	30.53	47.14	53.91	3.22
4	Low density lipoproteine cholesterol (LDL-C)	103.63	128.70	146.98	103.68	132.70	149.68	0.47
5	High density lipoproteine lecithin (HDL-Lec)	27.84	43.16	49.12	57.15	64.85	72.13	5.44
6	Apolipoprotein A1 (Apo A1)	70.40	81.20	86.99	109.33	119.00	140.50	6.63
7	Apolipoprotein A2 (Apo A2)	27.14	32.19	36.07	38.58	50.60	56.55	4.93
8	Apolipoprotein B (Apo B)	74.80	86.06	95.27	84.45	98.80	114.95	2.44
9	Triglycerides (TG)	103.25	135.00	167.25	88.50	125.00	156.00	0.87
10	Total cholesterol (TC)	161.75	185.00	215.50	169.25	200.00	228.50	1.22
11	Apo A1/Apo A2	2.31	2.54	2.87	2.34	2.66	2.86	0.91
12	Apo A1/HDL-C	2.35	2.81	3.03	2.61	2.98	3.27	2.01
13	Apo A2/HDL-C	0.88	1.08	1.30	0.99	1.12	1.25	0.60
14	Body weight index (BWI)	17944	18845	19776	17773	18635	19092	0.58
15	Systolic blood pressure (syst. BP)	130	150	170	121	140	150	1.84

[a] The distributions have been compared by a Wilcoxon U test. z is the value of the rank sum transformed into normal distribution.

information for an effective representation of elements of different classes and for an effective discrimination between elements of different classes.

In practical applications where only a finite, usually small, number of training data are available there are two reasons for variable selection (*21–25*):

(i) For finite sample sizes the inclusion of additional variables may not contribute additional information with respect to the discriminatory quality of a model.

(ii) Additional variables may increase the cost of data gathering and data processing.

Because of the high number of possible combinations of q variables for also relatively small q, an exhaustive search for an optimal solution of the evaluation criterion is nearly impossible. Therefore search strategies for suboptimal solutions have to be developed.

With regard to the stepwise maximum likelihood estimation of smoothing factors of the kernel functions [2], a stepwise "forward selection" procedure combined with a "backward elimination" procedure was developed:

0. Given q variables $x_{ij1}, \ldots, x_{ijq}$, $i = 1, \ldots, m$, $j = 1, \ldots, n_i$

1. Set model order $p = p_o$.

Usually $p_o = 0$. If $p_o > 0$, then the smoothing factors of $p_o - 1$ variables must be given.

2. $p: = p + 1$.

Add one of the $q - p + 1$ variables to the model a time. Estimate the smoothing factor of this variable according to the maximum likelihood criterion [4], when the smoothing factors of $p - 1$ variables are fixed. Compute the evaluation criteria.

3. Select the variable for which a certain evaluation criterion is optimal for models of order p (see next section).

4. Is the improvement of the model evaluation criterion according to some defined stopping rule substantial?

Yes: if $p < q$ then go to 2, else stop.

No: should the model order be increased by one?

Yes: select one variable

No: if $p = 1$ then stop, else start the backward elimination procedure; $p: = p - 1$, go to 5;

5. Every variable of the model of order p, which has been selected in step 3 is now substituted by the remaining $q - p$ variables. The smoothing factor of the new variable is estimated according to the maximum likelihood criterion [4] or, to save computing time, the value of the smoothing factor which was found in step 2 for model order p is used. Compute the evaluation criteria.

6. Is there a substantial improvement of the model evaluation criterion?

Yes: The model of order p with the best evaluation criterion is chosen to be the best for this model order. Go to 2.

No: Stop.

TABLE II

SMOOTHING FACTORS (h_{kk}^2) AND SOME EVALUATION CRITERIA FOR A MODEL OF ORDER $p = 1$ FOR $q = 15$ VARIABLES[a]

$X(k)$ $k =$	h_{kk}^2	tER	aER	COST	Q_1	Q_2	Q_5	Q_7
1	0.774	34.21	32.89	26	0.2556	0.7016	0.226	0.165
2	0.540	40.79	39.47	31	0.3171	0.7245	0.153	0.086
3	0.559	28.95	28.95	22	0.3680	0.7624	0.078	0.245
4	0.570	55.26	46.05	42	0.3042	0.7182	0.455	0.050
5	0.774	15.79	15.79	12	0.2245	0.6895	4.005	0.784
6	0.924	10.53	10.53	8	0.2120	0.6832	13.162	1.605
7	0.873	19.74	17.10	15	0.2390	0.6988	0.623	0.643
8	0.673	35.53	32.89	27	0.2819	0.7124	0.023	0.054
9	0.565	82.90	52.63	63	0.3390	0.7318	−0.302	−0.052
10	0.572	42.11	39.47	32	0.2892	0.7141	0.059	0.044
11	0.822	39.47	34.21	30	0.2563	0.6983	0.764	0.051
12	0.574	46.05	42.10	35	0.3100	0.7182	0.451	0.070
13	0.774	60.53	48.68	46	0.2773	0.7074	−0.277	−0.045
14	0.575	44.74	40.79	34	0.2524	0.6951	−0.046	−0.019
15	0.375	57.90	28.95	44	0.5463	0.6607	9.546	0.827

[a] List of variables, see Table I.

By this procedure a suboptimal model according to some defined evaluation criterion is identified. The "backward elimination" process (steps 5, 6) after the "forward selection" process is necessary because of the possibility, that the k "best" variables must not be the "best" k variables.

Under the assumption, that the variables are uncorrelated, matrix H is a diagonal matrix. The smoothing factors for each single variable would be the same if they are estimated simultaneously for all variables or only one a time for a single variable. Under this assumption, the stepwise parameter estimation described above would not be necessary. For most practical problems this assumption is not valid and therefore one possibility for economical smoothing factor estimation is to estimate only one constant smoothing factor h_c (*8, 9*) for all p variables or to perform the stepwise maximum likelihood estimation of smoothing factors described above. This stepwise procedure may be called an "adaptive" procedure. An application of this procedure to medical data (section 4) has shown that the value of the smoothing factor of a variable depends on the other variables which are in the model (Table III).

MODEL EVALUATION

There is no definite criterion for model evaluation for stepwise variable identification in discriminant analysis. Different evaluation criteria may be stated for different problems and they may lead to different "suboptimal" models in stepwise procedures.

Most model evaluation criteria described in the literature (*18, 26–28*) reflect a particular characteristic of the underlaying distribution space of the data.

They are measures for class separability and the degree of overlap of different classes.

The selection of an evaluation criterion has to be closely linked to the aim of discriminant analysis. Two aims can be distinguished:

(i) descriptive discriminant analysis: to identify the substantial different features between classes

(ii) predictive discriminant analysis: to classify future observations on the basis of an identified allocation rule.

For stepwise discriminant analysis model evaluation criteria should be used, which can be compared for different model orders p and for which appropriate stopping rules can be defined. Furthermore a model evaluation criterion should allow to weight right and wrong classifications according to the importance of classifying an element of class d_i as an element of class d_j, $i, j = 1, \ldots, m$. To evaluate the goodness of fit, it should also be possible to define upper and lower bounds for a criterion.

In order to estimate unbiased values of the evaluation criteria, test data should be used which are independent from the training data. One way to achieve a separation between training and test data without sampling additional test data is the estimation of characteristic values of an observation by jackknife or bootstrap procedures (*17, 29, 30*).

One type of evaluation criteria of discriminant functions are measures of the distance between classes like Mahalanobis distance D^2, Wilk's Λ, the multiple correlation coefficient R^2, the C_p statistic or other distance measures closely related to the F statistic (*18, 31–33*). These criteria can be used to evaluate linear discriminant functions, where the assumption of multivariate normal distribution with equal variance–covariance matrices is valid.

For nonparametric discriminant analysis model evaluation criteria which are continuous or discret functions of the probability density functions of the classes are more useful than the distance measures mentioned above. A large number of probabilistic distance and dependence measures are described in the literature (*13, 18, 27, 28*). One of the most important distance measures is the Kolmogorov variational distance for two populations or the Kolmogorov dependence for m populations and related measures. They describe the amount of overlap of a posteriori probability functions. The computational work for estimation of these criteria becomes very high for multivariate distributions. Generalized distance measures for more than two groups ($m > 2$) are based on averages of pairwise distances. Therefore for $m > 2$ dependence measures are computationally less complex than distance measures.

For practical applications with multivariate distributions ($q > 1$) evaluation criteria which need less computational work are preferred (*27, 28*). An evaluation criterion, which is simple to understand and to compute, is the mean probability (Q_1) of elements $x_{ij}, j = 1, \ldots n_i$ of the classes d_i, $i = 1, \ldots, m$. Closely related to Q_1 are the logarithmic score (Q_2) and criteria which are analogous to the definition of entropy (Q_3) (*27, 28*). These evaluation criteria

Q_1, Q_2, Q_3 only consider the probability of an element x_{ij} belonging to class d_i, but they do not consider the probability of x_{ij} belonging to class d_k, $k \neq i$. The scaled quadratic evaluation criterion (Q_4) also takes into account the probability of an element x_{ij} belonging to class d_k, $k \neq i$ (27, 28).

The nonerror rate (NER) and the error rate (ER) are used most frequently in practical applications for discriminant function evaluation. For many practical applications a cost or loss function COST is used to weight the costs misclassification. For nonparametric discriminant analysis decision rules similar to Bayes minimal error or minimal risk decision rules (9) are used for classification.

An enormous disadvantage of evaluation criteria ER, NER, COST is that the probability of an element x_{ij} belonging to class d_k is reduced to two discrete states: belonging to class d_k or not. This dichotomization is a loss of information and therefore we propose evaluation criteria, which are based on the relation of probabilities:

$$Q_5 = \frac{1}{n} \sum_i \sum_j v_{ij} \qquad [6]$$

with

$$v_{ij} = \frac{\hat{f}(x_{ij}/d_i;H)}{\max\limits_{k \neq i} \hat{f}(x_{ij}/d_k;H)} - 1 \qquad \text{if } \hat{f}(x_{ij}/d_i;H) \geq \max_{i \neq k} \hat{f}(x_{ij}/d_k;H)$$

$$= \frac{\max\limits_{i \neq k} \hat{f}(x_{ij}/d_k;H)}{\hat{f}(x_{ij}/d_i;H)} + 1 \qquad \text{if } \hat{f}(x_{ij}/d_i;H) < \max_{i \neq k} \hat{f}(x_{ij}/d_i;H).$$

Evaluation criterion Q_5 contains information about the mean relationship of probability densities of elements x_{ij}, $i = 1, \ldots, m$, $j = 1, \ldots, n_i$ to be classified into class d_i, $i = 1, \ldots, m$, compared to the maximal probability density of x_{ij} belonging to a class $\neq i$.

Similar to Q_1 and Q_5 other evaluation criteria can be defined:

$$Q_6 = \frac{1}{n} \sum_i \sum_j \hat{f}(x_{ij}/d_i;H) - \max_{k \neq i} \hat{f}(x_{ij}/d_k;H) \qquad [7]$$

and

$$Q_7 = \frac{1}{n} \sum_i \sum_j \log(\hat{f}(x_{ij}/d_i;H)) - \log(\max_{k \neq i} \hat{f}(x_{ij}/d_k;H). \qquad [8]$$

Compared to evaluation criteria ER, NER, and COST criteria Q_5, Q_6, and Q_7 take into account the degree of certainty at which a decision is made. Furthermore a cost matrix may be used to weight classification and to determine the cost of a discriminant model.

A disadvantage of evaluation criteria Q_2, Q_5, and Q_7 is, that lower and/or upper limits have to be defined, otherwise single elements x_{ij} could lead to biased estimates of the scoring rule. For criteria Q_2 and Q_7 an upper limit for u, $u < \varepsilon$ for $\ln(u)$ has to be defined. For evaluation criterion Q_5 this limitations means

$$|v_{ij}| \leq v_{\max}.$$

To evaluate the quality of a discriminant function with regard to the application on future data sets, it is necessary to estimate values of the evaluation criteria from test samples which are independent of the training sample used for the estimation of the parameters of the discriminant function. Splitting the data into a training and learning sample would load to small sample sizes in many practical applications.

Therefore, jackknife or bootstrap procedures (*16, 17, 29*) are proposed. In this study jackknife procedures are used and the values of the evaluation criterion Q_i, $i = 1, \ldots, 7$ are estimated using jackknife probability densities at points x_{ij}, $i = 1, \ldots, m\, j = 1, \ldots, n_i$ [5].

Using jackknife probabilities one gets "true" estimates of the evaluation criteria, like the true error rate (tER), instead of their apparent estimates, like the apparent error rate (aER).

Usually, jackknife procedures need very much computational work. For nonparametric discriminant functions, which are based on estimated probability density functions, the estimation of jackknife probabilities [5] does not need additional computational work.

For automatic stepwise variable identification procedures, it must be possible to compare evaluation criteria for different model orders p and to define stopping rules. These conditions are fulfilled by ER, NER, and COST and therefore the "apparent" and "true" values of this criteria are very often used in automatic procedures. On the other hand, these criteria use only a small amount of the information content of the probability density functions $f(x_{ij}/d_k;H)$ and in many applications different models with the same model order may have the same error (ER) and nonerror rate (NER) and costs (COST) and it is not possible to identify one optimal model for this model order by these criteria.

Evaluation criterion Q_1, Q_2, Q_3, Q_6, Q_7, and, to a certain extent, also Q_4 show a strong dependency on the model order p which is due to the properties of multivariate probability density functions. Therefore, these criteria are not very well suited for automatic stepwise variable identification. One of the most appropriate criterion for automatic stepwise variable selection in discriminant analysis seems to be criterion Q_5. The ratio of probability densities at point x_{ij} in Eq. [6] is independent of the absolute value and therefore of the model order. The scale of Q_5 is fine enough to differentiate between models of the same model order.

APPLICATION

The methodology described in the preceding sections is illustrated by an application of retrospective identification of risk factors for stroke. Blood pressure, body weight index, and biochemical blood values have been determined for each person (Table I). In this study the analysis of the data of 45 male risk patients and 31 male control persons will be demonstrated. All the men are from the same region and are matched as a group of similar age. Some variables deviate from normal distribution and therefore the application of linear discriminant analysis would not lead to optimal classification results (*3*). Furthermore, the variance–covariance matrices of the groups are different.

As can be seen from the medians x and the interquartile ranges ($I_{25} - I_{75}$), there is a high amount of overlap of the variables of the risk and of the control group (Table I). A comparison of the distributions by a Wilcoxon U test shows significant differences between risk and control group for some variables (*34*).

The application of stepwise linear discriminant analysis to these data has permitted the reduction of the apparent error rate to rather low values: ER = 2.6% for model order $p = 7$ (*35*). We will show that the application of procedures described in the preceding sections on the same data permits a further decrease of the error rate. Starting with model order $p = 1$ for $q = 15$ variables the maximum likelihood estimation [4] using jackknife probabilities [5] gives smoothing factors h_{ii}^2 in the range 0.375 to 1.070 in this example. Using the minimal true error rate (tER) as model evaluation criterion, the model for $p = 1$ is optimal for variable $x(6)$ (Table II). In this example no further model of order $p = 1$ with an error rate ER less equal 10.53% has been found. For model order $p = 2$ [4] models with the same error rate ER = 0.079 have been found. For example, variable $x(15)$ was chosen because, for this variable Q_5 [6] is maximal. An increase of the model order from $p = 2$ to $p = 3$ does not decrease the true error rate and therefore the automatic forward selection in the program stops. To get information about the possibility to reduce the true error rate for a model of order $p > 3$, additional variables up to model order $p = 6$ have been added. There was only a small decrease of the true error rate to ER = 0.079 for $p = 4$ whereas the apparent error rate decreased to ER = 0.0 for $p = 6$ and Q_5 increased (Table III). A backward elimination process, where the variables $x(k)$, $k = 2, 6, 11, 12, 14, 15$ have been substituted by the remaining nine variables did not improve the results.

Some data of the variable identification process are shown in Table II. Besides the model evaluation criteria, the changes of the smoothing factors of the variables for different models are striking. In this example the problems of automatic variable identification using the true or apparent error rate become obvious. For one model order p more than one model may have the same error rate and other criteria have to be used to select a variable. Furthermore the error rate may reach its minimal value for a certain model order and will not decrease for a higher model order whereas other model evaluation criteria will improve. For the evaluation of COST the cost matrix

$$C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

was used and for the computation of Q_5 [6] an upper limit $v_{max} = 100$ was used in this study.

To examine the influence of the model evaluation criterion on the variable identification process after the identification procedure for minimization of the true error rate for $q = 15$ variables, an identification procedure for the maximization of the mean ratio of probability densities at points x_{ij}, $i = 1, 2, j = 1, \ldots, n_i$ was performed for $q = 5$ variables $x(k)$, $k = 5, 6, 7, 14, 15$. As a stopping rule a minimal increase of 10% of Q_5 by increasing the model order p by one is used, starting with $Q_5 = 1$ for $p = 1$. The best model is found for $p = 4$ (Table III). A further increase of Q_5 for $p = 5$ is not substantial and the backward elimination procedure did not help to identify a better model.

The model with the lowest true error rate tER = 0.079 is the one of model order $p = 4$ with the variables $x(k)$, $k = 2, 6, 12, 15$. The one with the lowest apparent error rate aER = 0.0 is the model of order $p = 6$ with the variable $x(k)$, $k = 2, 6, 11, 12, 14, 15$. The model with the "best" value of Q_5 is the model of order $p = 4$ with the variables $x(k)$, $k = 5, 6, 7, 15$. The true error rate (tER = 0.132) of this model is rather high.

A comparison of the true and apparent error rate (Table III) makes the bias of the apparent error rate obvious.

To examine the importance of the stepwise maximum likelihood estimation of smoothing factors for three different values of the smoothing factor h_{66} of variable $x(6)$, the smoothing factors of a second variable $x(5)$ and $x(7)$ have been estimated according to the maximum likelihood criterion [4] using jackknife probability densities [5] of a model of order $p = 2$ (Table IV). For different values of the smoothing factor of one variable, different values of the smoothing fact of of the other variable have been found. To evaluate the results of the maximum likelihood estimation, the log-likelihood function

$$\log L(H) = \sum_{i=1}^{m} \sum_{j=1}^{n_i} \log \hat{f}_{K,J}(x_{ij}/d_i;H) \qquad [9]$$

has been computed (Table IV). The maximum of $\log L(H)$ [9] was found for the smoothing factor h_{66}, which was the value of he maximum likelihood estimation. Considering the influence of the smoothing factor on the smoothness of the probability density, it is not surprising that Q_5 increases with increasing smoothing factor.

Discussion

The assumption of a single smoothing factor for all variables in a multivariate probability density function for discriminant analysis (*8*) seems to be a rather strong simplification. To avoid the complex problems of simultaneous solution of the maximum likelihood functions [4] for all smoothing factors of a $q \times q$ smoothing matrix H, uncorrelated kernel functions have been assumed and a stepwise estimation of smoothing factors combined with stepwise variable identification is proposed. The procedure described in this study for normal kernels can also be applied to other kernel types and other types of variables.

The application of these procedures shows that the smoothing factor of one variable is not constant for all models. It depends on the other variables in a model. An evaluation of the maximum likelihood estimation of smoothing factors for model order $p = 2$ by the log-likelihood function [9] shows that log L

TABLE III

SMOOTHING FACTORS AND MODEL EVALUATION CRITERIA FOR SOME MODELS EXAMINED BY THE STEPWISE VARIABLE SELECTION PROCEDURE FOR $q = 8$ VARIABLES FOR $n = 45$ RISK PERSONS AND $n = 31$ CONTROL PERSONS USING THE TRUE ERROR RATE (tER) AND THE RATIOS OF PROBABILITY DENSITIES AT POINT $x(Q_5)$ (*) AS EVALUATION CRITERION[a]

$X(k)$, $k =$	*	$p = 2$				*$p = 3$		*	$p = 4$		*	$p = 5$	$p = 6$
2					0.621				0.621	0.621		0.621	0.621
5						0.675		0.675			0.675		
6	0.925	0.925	0.925	0.925	0.925	0.925	0.925	0.925	0.925	0.925	0.925	0.925	0.925
7								0.623			0.623		
11		0.823					0.725			0.775		0.823	0.823
12			0.575						0.872			0.872	0.872
14				0.723							0.974		1.02
15	0.554				0.554	0.554	0.554	0.554	0.554	0.554	0.554	0.554	0.554
tER	0.0921	0.1053	0.0921	0.0921	0.0921	0.1447	0.1053	0.1316	0.0790	0.1184	0.1447	0.1053	0.1053
aER	0.0789	0.0789	0.0789	0.0921	0.0658	0.0789	0.0526	0.0526	0.0658	0.0395	0.0263	0.0263	0.0000
COST	7.00	8.00	7.00	7.00	7.00	11.00	8.00	10.00	6.00	9.00	11.00	8.00	8.00
Q_1	0.0628	0.0574	0.0702	0.0516	0.0189	0.0178	0.0175	0.0060	0.0053	0.0049	0.0014	0.0015	0.0003
Q_2	0.5600	0.5551	0.5664	0.5496	0.5186	0.5176	0.5173	0.5060	0.5053	0.5049	0.5014	0.5015	0.5003
Q_5	21.40	21.23	17.66	12.62	23.06	30.21	28.85	36.69	28.51	27.34	35.47	32.55	32.42
Q_7	1.86	1.96	1.70	1.50	1.62	1.95	2.06	1.80	1.43	1.58	1.16	1.23	0.62

[a] List of variables, see Table 1.

TABLE IV

EXAMINATION OF THE INFLUENCE OF THE SMOOTHING FACTOR OF ONE VARIABLE ON THE SMOOTHING FACTOR OF A SECOND VARIABLE[a]

h_{66}^2	0.30	0.925	2.00
$X(5)$: h_{55}^2	0.924	0.624	0.571
log L	−267.5	−253.0	−271.4
tER	10.53	14.47	14.47
Q_5	36.69	20.49	8.94
$X(7)$: h_{77}^2	0.624	0.673	0.675
log L	−260.4	−246.0	−268.2
tER	11.84	10.53	18.42
Q_5	45.05	20.73	10.71

Spearman's rank correlation coefficient

	Risk	Control
$r_S(X(6)X(5))$	0.2444	−0.1097
$r_S(X(6)X(7))$	0.3815	0.5616

[a] The smoothing factor of the first variable is fixed, the one of the second variable is estimated according to the maximum likelihood criterion. n = 45 risk persons, n = 31 control persons.

becomes a maximum if both smoothing factors are estimated by the maximum likelihood method.

Small changes of the smoothing factors (±0.05) for variables standardized to variance 1 would not substantially influence the results. Therefore ±0.05 was used as the accuracy of the estimation of smoothing factors.

The determination of a model evaluation criterion for a discriminant function depends on the problem to be analyzed and is to some extent subjective. The apparent error rate, which is mainly used in practical applications is a biased value of the true error rate. In linear discriminant analysis the computation of the true error rate is very complicated, because splitting of the data into training and test data makes the computation of the parameters of the discriminant function for all training data necessary. For nonparametric discriminant analysis, the computation of the true error rate can be performed very easily using jackknife procedures. But for automatic variable identification, the true and also the apparent error rate are very rough criteria and may lead to many problems. Therefore, other criteria, like distance functions for multinormal data, or criteria which describe the amount of overlap and the relation of the security of a decision seem to be better qualified for automatic variable identification. Different model evaluation criteria may lead to different "suboptimal" models and therefore the criteria should be chosen very carefully and more than one criterion should be considered simultaneously for model evaluation.

The application of the stepwise variable identification procedure to medical decision problems shows the possibility of reduction of the number of variables without loss of discriminatory power.

ACKNOWLEDGMENTS

I thank Professor T. Kenner for many stimulating discussions, and Professor G. Kostner for the data.

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